Unit 3 - Impact of IT
Assignment 2

Introduction

Scenario / Vocational Context: *You work for an ICT Consultancy Company called iTech, the Company is involved in supporting all types of organisations from big companies to small charities with their ICT solutions. As the Consultancy is fairly new, you have been tasked with producing a range of marketing and promotional documents, the documents will be used to explain to organisations the reasons they may need to upgrade their ICT and the potential consequences of not doing so.*

Task 1

(U4:P1) - Explain the reasons for upgrading IT systems in an organisation

External pressure from regulators and legal pressure from specific laws

Improvement of external environmental issues add to the need for change which are economic and legislative, and Organizations revolutionize when regulatory and legal frameworks are changed. The internal policy is dependent on how the organization understands the legislation or regulatory frameworks.

For example, the use of P2P file sharing, which are programs that are shared between users when they don't own a file or program but pay a minor fee, has been prohibited in a few school and companies, even if they are legal. Even to this date, various changes in legislation are taking place, such as the raising of the minimum wage since October 2010, which had a benefit to the finances of few business in the present economic state.

Other policies such as PAT, which is to make sure that the equipment is frequently checking for safety and compliance with the Data Protection Act of 1998, are placed on a legal basis. The Display Screen Regulations of 1992, Computer Misuse Act of 1990 and Disability Discrimination Act of 2005 are all acts in which it requires the company to provide proper equipment for employees who are disabled or require help to complete their jobs, such as a trackball for a person with arthritis or wrist rests to help avoid repetitive strain injury.

These are legislations which will cause an impact on company policies. Also, a company should have a policy where it's website can be accessed by disabled people and this is a legal requirements in countries such as the USA.

Understanding Competitors, learning from them and evolving your business

One of the main reasons that any business chooses to update their systems is to keep up with companies similar to their own and outperform them.

E-commerce is a major factor in a business's success therefore if their IT systems do not perform well, it can impact the company causing them to lose revenue. In some situations, a company's competitor would have had updated their website previously and gained a larger customer base.

This would be a problem if the company's current website isn't sufficient and often causing customers that shop in the store not to shop online, but instead would go to another company's website.

Also, by using e-commerce, global companies face high potential for competition. This causes organizations operating online to gain business from more traditional traders.

For example, Tesco started online grocery shopping which quickly took business from its competitors. Other supermarkets would have lost more business if they had not followed. When the email was created, it had online competition and was seen as a intimidation on the Royal Mail. However, to this date, email has not cause the termination of the traditional postal service.

Growth and development of a Business to become globalised

Upgrading the IT systems in a business can really help develop the business to the modern world a lot more and the globalization of the business itself.

Globalisation means when there are developments in the technology that increases the access to the internet, email and video conferencing, which is helpful for organizations to find suppliers or distributors anywhere in the world.

The same method also applies to rival organizations which cause more competition and loss of business. This is why organisations need to make sure that they utilize the advantages of technology available in the most successful way, to decrease costs and increase efficiency without reducing quality.

Improved efficiency of processes and the benefits of better customer service

The customer is undoubtedly the most important feature in an organisation's operation. The customer will be satisfied if the product is manufactured and delivered in a lesser lead time without depleting its quality.

Also, the customer is expected to return the product if the organisation passes on any cost savings. This is far more powerful marketing than any other and it is usually free. Customer service is the most crucial aim for organisations and increasing productivity,

For example, some companies make their websites the core focus when in development so it can be easier for their customers to use and to provide better customer service and to get more people shopping from them online.

With an old website, the company might have complaints that it is difficult to find and purchase items, so this should be the key issue that they wanted to address while developing the website. Making the website more user friendly will encourage people to use the site to buy their products. The addition of features such as filters, outfit guides and what's in season helps their customers find what they went quickly and easily, while maintaining high standards of customer service.

The designing of a new website will make it more attractive to customers and combined with new, better features customers are more likely to reuse the website or recommend it.

Recommendations from online users will boost the presence of company, which in turn will improve the sales and profit of the organisation. Through this the company can see how the website is successful and what faults there are in order to improve it and give the customer a better experience. The company can develop new features to draw in more customers and increase the productivity of the company.

Management Information Systems (MIS) (How data is handled)

Upgrading the MIS for a growing business is a very significant upgrade due to the mass amount of information that would flow into a growing business.

For example, the biggest online shopping company, Amazon, must have a lot of information within their database. The reason for this is because they have to cover many different products that they are selling within their website, such as the category of the product, whether it is used or new, etc.

Another significant reason of upgrading the MIS to a much newer and efficient one is due to the vulnerability the unupgraded MIS. The reason for this is due to the fact that it could be easily intercepted by an anonymous user, stealing the highly valuable information which could cause the downfall or bad reputation of the business.

(U4:M1) - Examine why an organisation needs to keep pace with IT developments

How investing in technology can increase profits and reduce expenses in the long-term

Investing in technology gave provide the business with many benefits such as improved speed, the ease of sharing and storing information and a decrease in human error through automation add up to a reduction in costs and an increase in revenue.

Since nearly every business rely on the internet for part of its operations, with some operating entirely online, high-speed internet technology is an important investment for any company. With satellite internet speeds increasing all the time, there's no limit to the potential for this technology.

Business can benefit financially from faster internet by saving time on all internet-based operations and reducing downtime for their website. With the downtime of their website being less frequent, they can produce a lot more profit without being obstructed from the downtime.

In general, Investing in the technology can help result in a financial gain for the company, a higher rank in the competitive market of retail and a boost in the overall productivity of the company.

The benefits of reinvesting profits into IT upgrades and improvements

With technology, many businesses of any size can increase their productivity, which translates into a lower cost structure. However, there are some other benefits that can help the businesses as well.

The speed and time are a big increase for a business because it makes it more swift and agile. What I mean by this is that before technology, email didn't exist, and the only way to communicate with someone else in another geographical location was by postal service.So the farther away your client or supplier, the longer the communication took.

Today, with the advent of information technology, information is shared at an astounding rate. This saves time, offering the ability to make decisions faster.

Easier Storage Technology eliminates the need for double or triple entry systems and reduces the need to file large amounts of paperwork. With the technology today, contracts and customer information can be stored in virtual data warehouses and accessed in minutes or even seconds, which cuts down on the need to purchase or rent storage space.

Technology can also improve the sharing of information. What I mean by this is that it allows information, whether written or broadcast, within a business to be shared more quickly and with fewer resources.

For example, Marketing can be accomplished by placing ads that reach millions of ready buyers on the Internet or through social networking sites. E-learning and other forms of online training have reshaped the readiness of the average small business workforce as employees can listen to classroom lectures and share ideas with classmates from the comfort of their home or office.

Technology can also improve with automation. What I mean by this is that this allows businesses to automate certain functions that historically have required the need to hire an employee.

For example, bookkeeping functions now can be handled by software applications such as Quicken and Quickbooks. The sales function is automated through contact management sites such as SalesForce. This gives the business owner the ability to focus on strategy and cut down on labor expenses.

The benefits of being able to turn company data into information that is useful

Thanks to the internet, business have a lot more data than ever at their fingertips. Arguably, knowledge is money and a business can recognize the potential of all this information. With the right data, a business can pinpoint their target audience, create accurate business projections and make informed decisions on all front.

According to online studies, about 40 percent of business believe that big data is linked to increased profitability, though the majority of businesses are intimidated by the barriers to data collection and storage. Fortunately, technology like Hadoop offers an answer. This open source software framework allows for massive data storage on a small budget.

The company data that is contained in massive online companies such as Facebook and Amazon, give a great amount of information about users on the internet. The information that the companies have can be used to tailor the needs of each of the customers, giving the customer a great amount of satisfaction and loyalty to those businesses.

This means that the customer would most likely use word of mouth or other forms telling others that it is a very good company, which attracts a lot of people to use the product or service that the company has created. This will cause a chain reaction of customers, meaning that the company will grow strong with loyal customers and profits.

Why up to date and modern IT systems that support improved customer service are a worthwhile investment

There’s no question that social media marketing is a popular and effective way to connect with customers and increase profits.

The problem is that a strong social media presence requires a major time commitment. Customers respond best to brands that post regularly and stay active on sites like Facebook and Twitter daily. Companies can increase their efficiency on social media sites with automated tools that take much of the legwork out of this type of marketing, freeing staff members up for other campaigns.

The goal of the business in terms of its customer interactions is to generate loyalty. Offering quality products and services and to be responsive to the customers is a very good way of getting loyalty from customers.

However, as new technologies have come to the market to make it easier for businesses to provide customer service, they may also be increasing the number of channels through which the business interacts with customers and the complexity of those interactions.

To improve the quality of the customer service, the business would need to focus and improve on many factors, such as data management and analytics (Using data collected from customer to analyze their preferences), insight-driven marketing (Gaining insights into your business from customer data so you can more effectively target marketing), marketing automation (Streamlining and automating business processes to improve efficiency and keep costs low), self-service optimization (Finding ways for customers to interact with your business when they want) and workforce effectiveness (Encouraging your staff to embrace new ways improving customer treatment by providing tools and training to deliver better service).

Task 2

(U3:P3) – Explain the issues related to the use of information

Legal issues related to the use and management of Data and Information. Making reference to specific laws and legislation

There are a few legal issues which were introduced to protect the usase and the data stored within the computer system. For example, the Data Protection Act of 1998 and the Computer Misuse Act of 1990.

The Data Protection Act, which was introduced in 1998, was created to ensure that the personal information and data within a computer system is handled correctly. What I mean by this is that this act stops the distribution and improper use of a person's data by ensuring that all companies registered with the Data Protection Act follow strict guidelines. The guidelines have clear statements on how to carry out this act.

For example, some of the guidelines that the act state that: all the information kept by an organisation must be fairly and lawfully processed; The informations kept can only be processed for the purpose it is being used for; Only a limited amount of information (not excessive) can be stored and it must be adequate and relevant.

The information about the person is accurate and has only been recently collected; It can only be stored for a limited amount of time; It has to be processed in accordance with the person's legal rights; It has to be stored securely so that only the company can access the information

and that the information must not be distributed to other countries without the correct amount of protection.

The reason that the Data Protection Act may affect the company when using secondary information and data from the internet is that, if the company uses information from the internet, it could breach the Data Protection Act meaning the company would get in trouble and could face serious consequences. However, in 2006, there were modifications made to this act in the Police and Justice Act to make this act more secure.

The Computer Misuse Act, which was introduced in 1990, covers the offences that people make when using computer which can cause a breach of the security of information. The information consist of unauthorised access of another computer - typically using login details of another person, unauthorised access to compute hardware with intent to commit an offence of a serious nature and unauthorised manipulation of data or information without the permission of the owner.

The reason the Computer Misuse Act may affect the company is that if the virtual team attempts to impede the other companies so that they win the competition.

For example, if there was an opposing company leaves their login details for the server vulnerable to another opposing company, and that the vulnerable information was to be breached by the opposing company and used to change the operation settings of the server, they would be breaching two of the conditions of the Computer Misuse Act and could of faced serious legal implications.

Ethical issues related to staff, codes of practice and information ownership

Usually, Companies have a code of practice which ensures that all people within the company use the facilities available to them properly. It defines the uses of the different facilities, for example the use of email and the internet.

In a company, there is often a code of practice for using emails as using it incorrectly can cause implications for the company. Sending emails that contain inappropriate content or are deemed threatening or harassing is a banned practice in a company as represents the company in a negative way.

Another code of practice for the use of emails would be the banning of sending spam/multiple emails as this can be considered harassment and falls under the criteria of the first point mentioned. To stop people from doing this, most companies limit the amount of emails a person can send for private purposes.

Like the use of email, companies have a code of practice for the use of the internet. This is to stop employees from accessing websites that are inappropriate, such as gambling and

inappropriate websites. Companies enforce this by using filtration software and limiting internet use for personal use.

The implementation of codes of practice stops Code from using the company's facilities incorrectly. If they do not abide by the code of practice, it damages the company's reputation and can weaken the partnership with external companies or organisations. It can also affect the company's position in the competition and cause them to be disqualified.

An example of a breach of the code of practice would be if one of the teams spammed the Cisco email to get a response from the mentor instead of waiting for a reply.

Organisational policies affect how a company handles information. There are two types of organisation: centralised and decentralised. The type of organisation will affect how information is stored and protected.

Hierarchical organisations and companies distribute information on a need to know basis. This means a policy will be developed that only allows the necessary parties to view information, in which it is managed by admin who can change who can view the information.

Decentralised organisations have parts of the organisation in different geographical locations. This means that file security would be more relaxed when accessing information within one site, but more strict when accessing data from a different site.

An organisation policy may affect our company in many ways as we have a hierarchal structure within the company. It can stop the different sub-teams and people lower down in the hierarchy from accessing information without permission, which can be both a positive and negative attribute.

For example, if there is private information that only the program and project managers need to know, then the organisation policy implemented on a need to know basis is a good for managing information, if everyone could access the information it could cause more issues within the group.

A negative effect of this type of organisational policy would be that sharing information between the teams would be more difficult. Organisational policies can also aid information flow in the group.

<u>Operational Issues related to an Organisation's systems and policies</u>

Security of information means that information is secure and is not susceptible to unauthorised access and data manipulation. The IT department in most companies are responsible for this, however it is a company's manager that is responsible for deciding who can view and use or change the information.

What I mean by this is that the manager can develop a security management plan to ensure that information on a network is secure and the security of information can affect the company in multiple ways. If information that is stored for the company isn't secured, it could be accessed by the other companies and used against the team.

An example of this would be where the physical security project manager was away and the rest of the team needed the designs. The project manager posted images of the designs on Facebook where people from the opposing companies could view the designs.

Another example would be when the team put some of the files on a file sharing program where the members and teams within the company could access them.

Also, a company should make frequent backups of information so that it is not lost if the network the information is stored on fails. This means any information stored on the network is copied to prevent any issues if the original copy is lost.

Backups can be done for all of the information in a network/system or just of information that has been modified since the last full backup. Tests should be run to ensure that if the network does fail, it can be fully restored/recovered using the last full memory backup.

By backing up information, the company is protected if the school network fails. Any information stored on the laptops or computers used in the company would be lost if the network failed, so by backing up the information, the project is not affected. If the different teams do not backup their information, it can mean that they have to redo all the work which means they would fall behind in the project.

For example, if a team within a business working on a prototype project and that the team did have the designs for the prototype and they did not have a backup copy, the designs would be very vulnerable. The reason for this is due to the chance of the design being lost and never found.

If the team does upload the prototype onto a backup, the issue would be solved because the designs were in a secure location and the copy that was uploaded onto a cloud storage device would have access anytime on a variety of devices. If the issue arose again, there will be a permanent backup copy available for them to use.

(U3:M2) – Assess how issues related to the use of information affect an organisation

What happens if they get it wrong?

If a large company like Amazon were to get either the legal, ethical or operational issues wrong, it could have a massive impact to the company either legally, financially or by customers.

With the legal issues, there could be serious consequences depending on what type of legal issues, whether it is Data Protection Act or the Computer Misuse Act.

If Amazon were to breach the Data Protection Act, it would have to face the many legal consequences that might arise towards the company. However, not only would it be impacted by the legal point of view, but also the financial and customer loyalty.

For example, if there was an opposing company leaves their login details for the server vulnerable to Amazon, and that the vulnerable information was to be breached by Amazon and used to change the operation settings of the server, they would be breaching two of the conditions of the Computer Misuse Act and could of faced serious legal implications.

If this scenario was to occur, many funds from the company would need to be taken from the law because they would be sued. Not only that, but the customers would also see the company as a dishonourable company towards its competitors and may not use the company again because of the loss of customer loyalty, reducing the profits for the company.

With the ethical issues, Amazon would have to face a lot of issues with human rights groups and, in extreme cases, the law. Since Amazon have warehouses and offices in which people work in and maintain the company, there are many ethical issues that can occur.

For example, it is possible that Amazon would have employees in their company which are unable to do certain activities, so the company would need to invest on compensating the disabilities of the employees.

However, if the company didn't focus on the ethics of it's employees, it would face many problems with the employees and the human rights groups. This could make employees either quit the job, which would leave empty spaces or cause a lot of the employees to go on strike, giving the company bad publicity. Having bad publicity can influence the customers who notice this to be disloyal to the company because of it's lack of ethics.

Also, if the employees of the warehouses didn't get a good amount of ethical improvements, the employees may face serious injuries or other extreme cases. This would cause the law to enforce Amazon, which would make it pay its fund to compensate for the injuries caused to the employees who were unable to do a task properly because ethical improvements weren't invented.

With the operational issues, it can lose a significant amount of data and information. This would be very hard hitting for a company like Amazon because it is the large amounts of data that makes Amazon such a large company.

For example, if a team within Amazon is looking after a server and the security and that the team did have the server in working order but they did not check the security, the server would be very vulnerable. The reason for this is due to the chance of the server being breached and the information that it leads access to would be breached.

This that Amazon would face is that the information and data would be intercepted and stolen, which can lead to the data and information to be leaked or even stolen and never found. THis would be catastrophic for Amazon because their data and information is one of the most significant priorities of Amazon, and loosing it would make Amazon very weak compared to other competing companies.

Another problem that this would cause would also be the server being shut down, causing their website to be in downtime. Which would reduce the amount of products sold because it is the only method in which Amazons sells their goods, causing their profits to drop due to the downtime.

In conclusion, if Amazon were to ignore the legal, ethical and operational issues, it would impact the company heavily financially and on the customers. Due to the costs of the enforcement and the loss of customers who notice that the company isn't a loyal company to be with.

What is their business system? How will it be affected?

Amazon's business system is that it is an online retail, which is basically a business which includes products sold online via a website. Amazon claims to have "Earth's Biggest Selection" of products available through its family of websites, sold at the lowest cost at a small profit.

However, Amazon doesn't stock everything that is sold through its website. Another part of its retail strategy is to serve as the channel for other retailers to sell their products and taking a cut of every purchase. Amazon maintains its status as a destination website, but does not have to maintain inventory on slower-selling products.

This business system can be affected because it is an online retail company, making it very vulnerable if it wasn't secured properly. What I mean by this is that Amazon sells its products online, so it needs its customers to visit the website in order to buy products and give money towards the business.

If an issue (especially an operational issue) were to occur, it may cause Amazon to lose access to the website to the customers or even vital customer information to be leaked or erased. If the website were to be shut down temporarily, it would reduce the flow of customers towards the company's website.

This would instantly stop the customer flow, which means that the products cannot be sold. This would cause the company to lose a significant amount of profit, which can even have an impact of the biggest of companies depending on how long the profits loss goes on for.

In the worst case scenario, Amazon's website may be shut down permanently and erased completely. This means that the website would have to be built from scratch, which can be very time consuming.

Time is very important to a company such as Amazon because it needs to make money out of the most time possible, and if that time were to be cut because of the remaking of the website, the company may be on a profit loss and even debt, risking it to be bankrupt.

If the information was leaked, this would cause Amazon to lose the vital information which it uses to satisfy customer needs.

Rival companies may take the leaked information and use it to their advantage so that the customers that used to be on Amazon would have a database on the rival company, which can attract them if they were to advertise on email or other forms of communication. Which could cause the risk of Amazon losing potential customers.

If the information were to be erased or stolen from Amazon, the company would be having a devastating time on trying to recollect the information for customers because they need to retrace all of the Amazon users. This could impact the sales that amazon does because there won't be products that interest the user on their website due to the lack of information.

Since the data for the customers was either stolen, it would mean that the company that stole the data and managed to not get noticed, it would mean that the company would use the customer data to advertise themselves and manipulate users to visit their company. Amazon wouldn't be able to combat this because they wouldn't be able to trace the company and wouldn't have the customer data to get the customers back and satisfied with the website.

In conclusion, Amazon's business system is set on a website where it sells goods towards customers, which means that the customer would need to have internet access to their servers that contain the website in order to buy products. This type of business system is very modern but vulnerable to interceptor compared to a traditional retail shop.

The vulnerability can consist of data theft, data leaking or server downtime. However, these are very few of the many problems that can occur to an online retail business system like Amazon.

<u>What data and information are they 'managing'?</u>

Amazon is managing data and information which is known as "Big Data". Big data can be described in terms of data management that, due to increasing volume, velocity and variety of

data, can't be solved with traditional databases. While there are plenty of definitions for big data, most of them include the concept of what's commonly known as the "three V's" of big data.

Which consists of: volume: Ranges from terabytes to petabytes of data; Variety: Includes data from a wide range of sources and formats (e.g. web logs, social media interactions, ecommerce and online transactions, financial transactions, etc); Velocity: Increasingly, businesses have stringent requirements from the time data is generated, to the time actionable insights are delivered to the users. Therefore, data needs to be collected, stored, processed, and analyzed within relatively short windows, ranging from daily to real-time.

The data and information that Amazon contains is the customer click-stream data and historical purchase data of about 152 million customers and each user is shown in customized results on customized web pages.

Amazon uses the big data to offer a superb service to its customers. They can do this because they use all the data they have collected from their customers to build and constantly improve the relationship with its customers.

However, Amazon is expanding its usage of big data since it notices that the competition is nearing closer. As such, Amazon added remote computing services, via Amazon Web Services, to their already colossal product and service offering.

The Amazon Web Services was launched in 2002, however only recently they added big data services and they now offer tools to support data and information collection, data and information storage, data computation along with data collaboration and data/information sharing, which are all available in the cloud.

In conclusion, Amazon is holding a lot of the data from over 152 million Amazon users so that it can be used to track what the customers are doing and buying. The reason that this is done is so that the website can be automatically tailored for the specific customer needs such as their recent purchases, the most sold item, the items that are discounted and the items that are related to what the customer buys.